Q1 - 2024 (01 Feb Shift 1)

For $0<\theta<\pi/2$, if the eccentricity of the hyperbola $x^2-y^2\operatorname{cosec}^2\theta=5$ is $\sqrt{7}$ times eccentricity of the ellipse $x^2\operatorname{cosec}^2\theta+y^2=5$, then the value of θ is :

(1) $\frac{\pi}{6}$

(2) $\frac{5\pi}{12}$

(3) $\frac{\pi}{3}$

(4) $\frac{\pi}{4}$

Q2 - 2024 (01 Feb Shift 1)

Let $\frac{\mathrm{x}^2}{\mathrm{a}^2}+\frac{\mathrm{y}^2}{\mathrm{b}^2}=1, \mathrm{a}>\mathrm{b}$ be an ellipse, whose eccentricity is $\frac{1}{\sqrt{2}}$ and the length of the latus rectum is $\sqrt{14}$. Then the square of the eccentricity of $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is :

(1) 3

(2) $7/2$

(3) $3/2$

(4) $5/2$

Q3 - 2024 (27 Jan Shift 2)

Let e_1 be the eccentricity of the hyperbola $\frac{\mathrm{x}^2}{16}-\frac{\mathrm{y}^2}{9}=1$ and e_2 be the eccentricity of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, a>b$, which passes through the foci of the hyperbola. If $\mathrm{e}_1\mathrm{e}_2=1$, then the length of the chord of the ellipse parallel to the x-axis and passing through $(0,2)$ is :

(1) $4\sqrt{5}$

(2) $\frac{8\sqrt{5}}{3}$

(3) $\frac{10\sqrt{5}}{3}$

(4) $3\sqrt{5}$

Q4 - 2024 (30 Jan Shift 1)

Let the latus rectum of the hyperbola $\frac{x^2}{9} - \frac{y^2}{b^2} = 1$ subtend an angle of $\frac{\pi}{3}$ at the centre of the hyperbola. If b^2 is equal to $\frac{l}{m}(1 + \sqrt{n})$, where l and m are co-prime numbers, then $l^2 + m^2 + n^2$ is equal to ______

Q5 - 2024 (30 Jan Shift 2)

Let P be a point on the hyperbola $H : \frac{x^2}{9} - \frac{y^2}{4} = 1$, in the first quadrant such that the area of triangle formed by P and the two foci of H is $2\sqrt{13}$. Then, the square of the distance of P from the origin is

(1) 18

(2) 26

(3) 22

(4) 20

Q6 - 2024 (31 Jan Shift 1)

If the foci of a hyperbola are same as that of the ellipse $\frac{x^2}{9} + \frac{y^2}{25} = 1$ and the eccentricity of the hyperbola is $\frac{15}{8}$ times the eccentricity of the ellipse, then the smaller focal distance of the point $\left(\sqrt{2}, \frac{14}{3}\sqrt{\frac{2}{5}}\right)$ on the hyperbola, is equal to

(1) $7\sqrt{\frac{2}{5}} - \frac{8}{3}$

(2) $14\sqrt{\frac{2}{5}} - \frac{4}{3}$

(3) $14\sqrt{\frac{2}{5}} - \frac{16}{3}$

(4) $7\sqrt{\frac{2}{5}} + \frac{8}{3}$

Q7 - 2024 (31 Jan Shift 1)

Let the foci and length of the latus rectum of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$ be $(\pm 5, 0)$ and $\sqrt{50}$, respectively. Then, the square of the eccentricity of the hyperbola $\frac{x^2}{b^2} - \frac{y^2}{a^2 b^2} = 1$ equals

Answer Key

Q1 (3) **Q2** (3) **Q3** (3) **Q4** (182)

Q5 (3) **Q6** (1) **Q7** (51)

Q1

$$e_h = \sqrt{1+\sin^2\theta}$$
$$e_c = \sqrt{1-\sin^2\theta}$$
$$e_h = \sqrt{7}e_c$$
$$1+\sin^2\theta = 7\left(1-\sin^2\theta\right)$$
$$\sin^2\theta = \frac{6}{8} = \frac{3}{4}$$
$$\sin\theta = \frac{\sqrt{3}}{2}$$
$$\theta = \frac{\pi}{3}$$

Q2

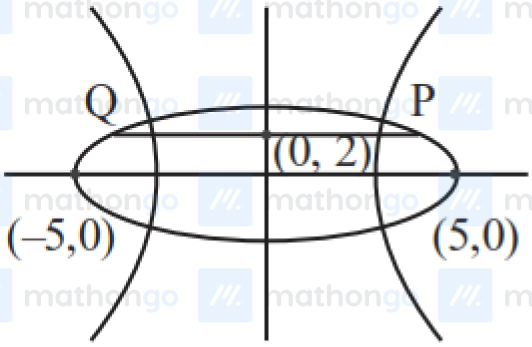
$$e = \frac{1}{\sqrt{2}} = \sqrt{1-\frac{b^2}{a^2}} \Rightarrow \frac{1}{2} = 1-\frac{b^2}{a^2}$$
$$\frac{2b^2}{a} = 14$$
$$e_H = \sqrt{1+\frac{b^2}{a^2}} = \sqrt{1+\frac{1}{2}} = \sqrt{\frac{3}{2}}$$
$$\left(e_H\right)^2 = \frac{3}{2}$$

Q3

$$H : \frac{x^2}{16} - \frac{y^2}{9} = 1 \quad e_1 = \frac{5}{4}$$
$$\therefore e_1 e_2 = 1 \Rightarrow e_2 = \frac{4}{5}$$

Also, ellipse is passing through $(\pm 5, 0)$

$$\therefore a = 5 \text{ and } b = 3$$
$$E : \frac{x^2}{25} + \frac{y^2}{9} = 1$$

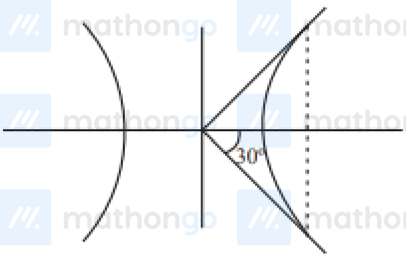


End point of chord are $\left(\pm\frac{5\sqrt{5}}{3}, 2\right)$

$\therefore L_{PQ} = \frac{10\sqrt{5}}{3}$

Q4

LR subtends 60° at centre



$\Rightarrow \tan 30^{\circ} = \frac{b^2/a}{ae} = \frac{b^2}{a^2e} = \frac{1}{\sqrt{3}}$

$\Rightarrow e = \frac{\sqrt{3}\,b^2}{9}$

Also, $e^2 = 1 + \frac{b^2}{9} \Rightarrow 1 + \frac{b^2}{9} = \frac{3\,b^4}{81}$

$\Rightarrow b^4 = 3\,b^2 + 27$

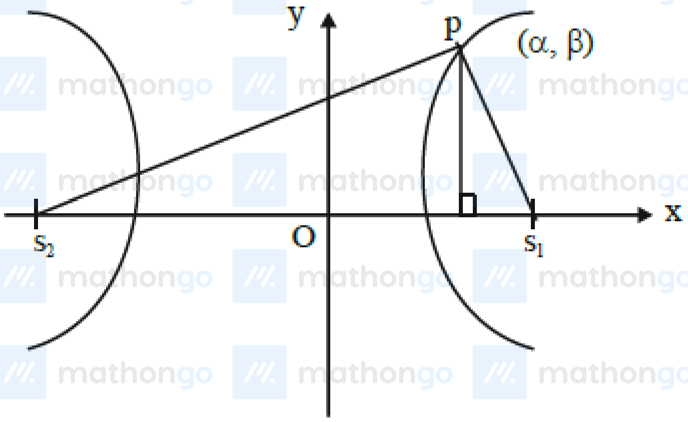
$\Rightarrow b^4 - 3\,b^2 - 27 = 0$

$\Rightarrow b^2 = \frac{3}{2}(1 + \sqrt{13})$

$\Rightarrow \ell = 3,\ m = 2, n = 13$

$\Rightarrow \ell^2 + m^2 + n^2 = 182$

Q5



$$\frac{x^2}{9}-\frac{y^2}{4}=1$$
$a^2=9, b^2=4$

$$b^2=a^2\left(e^2-1\right) \Rightarrow e^2=1+\frac{b^2}{a^2}$$

$$e^2=1+\frac{4}{9}=\frac{13}{9}$$

$$e=\frac{\sqrt{13}}{3} \Rightarrow s_1 s_2=2 a e=2 \times 3 \times \sqrt{\frac{13}{3}}=2 \sqrt{13}$$

Area of $\Delta PS_1 S_2=\frac{1}{2} \times \beta \times s_1 S_2=2 \sqrt{13}$

$$\Rightarrow \frac{1}{2} \times \beta \times(2 \sqrt{13})=2 \sqrt{13} \Rightarrow \beta=2$$

$$\frac{\alpha^2}{9}-\frac{\beta^2}{4}=1 \Rightarrow \frac{\alpha^2}{9}-1=1 \Rightarrow \alpha^2=18 \Rightarrow \alpha=3 \sqrt{2}$$

Distance of P from origin $=\sqrt{\alpha^2+\beta^2}$

$=\sqrt{18+4}=\sqrt{22}$

Q6

$$\frac{x^2}{9}+\frac{y^2}{25}=1$$
$a=3, b=5$

$$e=\sqrt{1-\frac{9}{25}}=\frac{4}{5} \therefore \text{ foci }=(0, \pm b e)=(0, \pm 4)$$

$$\therefore e_H=\frac{4}{5} \times \frac{15}{8}=\frac{3}{2}$$

Let equation hyperbola

$$\frac{x^2}{A^2}-\frac{y^2}{B^2}=-1$$

$$\therefore B\cdot e_H=4 \quad \therefore B=\frac{8}{3}$$

$$\therefore A^2=B^2\left(e_H^2-1\right)=\frac{64}{9}\left(\frac{9}{4}-1\right)\therefore A^2=\frac{80}{9}$$

$$\therefore \frac{x^2}{\frac{80}{9}}-\frac{y^2}{\frac{64}{9}}=-1$$

Directrix : $y=\pm\frac{B}{e_H}=\pm\frac{16}{9}$

$$PS=e\cdot PM=\frac{3}{2}\left|\frac{14}{3}\cdot\sqrt{\frac{2}{5}}-\frac{16}{9}\right|$$

$$=7\sqrt{\frac{2}{5}}-\frac{8}{3}$$

Q7

$$\text{focii } \equiv(\pm 5,0);\frac{2b^2}{a}=\sqrt{50}$$

$$a=5 \quad b^2=\frac{5\sqrt{2}a}{2}$$

$$b^2=a^2\left(1-e^2\right)=\frac{5\sqrt{2}a}{2}$$

$$\Rightarrow a\left(1-e^2\right)=\frac{5\sqrt{2}}{2}$$

$$\Rightarrow \frac{5}{e}\left(1-e^2\right)=\frac{5\sqrt{2}}{2}$$

$$\Rightarrow \sqrt{2}-\sqrt{2}e^2=e$$

$$\Rightarrow \sqrt{2}e^2+e-\sqrt{2}=0$$

$$\Rightarrow \sqrt{2}e^2+2e-e-\sqrt{2}=0$$

$$\Rightarrow \sqrt{2}e(e+\sqrt{2})-1(1+\sqrt{2})=0$$

$$\Rightarrow (e+\sqrt{2})(\sqrt{2}e-1)=0$$

$$\therefore e\neq-\sqrt{2}; e=\frac{1}{\sqrt{2}}$$

$$\frac{x^2}{b^2}-\frac{y^2}{a^2\,b^2}=1 \quad a=5\sqrt{2}$$

$$b=5$$

$$a^2b^2=b^2\left(e_1^2-1\right)\Rightarrow e_1^2=51$$